GETTING A DWELLING TO 7 STAR

Minimum insulation values – To input as std.
R3.5 to ceilings
R1.5 to walls
R1.5 to Garage internal walls
R1.5 to timber sub floors, and upper floor areas that overhang ground floor.

Glazing values - Generic single glazed clear
ALM-001-01 Type A (Awning, Bifold, Casement)
ALM-002-01 Type B (Sliding, Fixed, Louvre)

Steps to Upgrade

1. Ceiling Insulation
   - Try R4.0, R5.0, R6.0 & Max Value R7.0
   - note: must ensure there is enough roof space to allow this (290mm thick) plus gap for ventilation so would need approx. 350mm
2. Wall Insulation
   - Try R2.0, R2.5 & Max Value R2.7 HD batt to external walls
   - Add R2.0 batt to internal walls between Un-Conditioned & Conditioned Zones
   - Can also request an air gap if this helps and if boundary setbacks allow this.
   - note: must ensure there is enough space to allow this R2.5 & R2.7 require 90mm
3. Roof Insulation
   - Try Reflective foil, R1.3 Blanket, Max Value R1.8 blanket **for metal roof only**.
   - note: requires min. 25mm gap between reflective foil and top of ceiling batt
4. Floor Insulation
   - Timber Floors try R2.0 Max Value R2.5
   - Concrete Floor try 20mm,30mm,40mm, Max Value 50mm of Polystyrene Extruded (rigid foam)
   - Upper floor areas directly above a Garage try R2.5 or Max Value R3.5
   - Upper floor areas that overhang ground floor try R2.5 or Max Value R3.5
5. Glazing
   - Use generic double glazed values first ALM-005-03 Type A (Awning, Bifold, Casement)
     ALM-006-03 Type B (Sliding, Fixed, Louvre)
   - Use custom values - if not specified in checklist or on plans try A&L, Bradnams, AWS , Dowell
   - Alum. Frame - Lowest U Value - 2.5. SHGC value will vary depending on climate. Please remember is specifying a low SHGC (under 0.40) this will require a darker tint and may not be appealing to the design and facade.
   - Timber or uPVC Frame - Lowest U Value - 1.9

If the above specs still do not achieve compliance, then we can look at the below and perhaps need client consultation.

If we are trying to reduce cooling loads

- Then looking at adding thermal mass into the building exposed slabs, reverse brick veneer, concrete block or double brick internal feature walls will all help.
- Colour, specific lighter colours (SA<0.40) to roof, walls, window/door frames
- Adding ceiling fans and increasing fan diameter to 1400mm
- Adding more cross ventilation, looking at window placement across rooms or increasing ventilation openings to windows/doors
- Increasing shading. Vertical adjustable shading is recommended to East and West orientations.
- Specify Alphafloor for timber sub floors.
- Reducing East and West Glazing will always help, sufficient Northern glazing with appropriate horizontal shading.

If we are trying to reduce heating loads

- Reduce glazed area, make windows/doors smaller
- Adding thermal mass into the building exposed slabs, reverse brick veneer, concrete block or double brick internal feature walls will all help however for heating loads we must be more diligent in the location of the thermal mass element, if it is not receiving good solar access then it may have a negative effect as it will suck heat out of a space. So ensure thermal mass is exposed to Direct sun
- Colour, specify darker colours (SA>0.60) to roof, walls, window/door frames
- Insulation between unconditioned and conditioned zones
- Increasing Living Zone spaces if possible as these have a better natural heating load vs Day-Time
- Specify Alphafloor for timber sub floors.
- Reducing East and West Glazing will always help, sufficient Northern glazing with appropriate horizontal shading. Reducing southern glazing will also help for heating loads.